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WEAR INDICATOR FOR A WATER-REPELLENT FINISH

Provisional patent specification

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FIELD

This invention relates to wear indicators for water-repellent finishes on textile articles.

BACKGROUND

Durable water-repellent finishes on garments, tents, covers and awnings wear away with abrasion, soiling and washing. Finishes free of fluorinated compounds generally wear away sooner. When the finish fails, the outer fabric absorbs water. The owner has no indication of this other than deliberately wetting the article and observing whether water beads.

SUMMARY OF THE INVENTION

In one aspect the invention provides: a wear indicator for a water-repellent finish on a textile article, comprising an indicator layer of textile that carries a water-repellent finish and is located where the article is subject to wear, and a marking beneath or within the indicator layer. The marking is concealed while the indicator layer is dry and is revealed when the indicator layer absorbs water. The marking is therefore revealed on exposure to water only after the finish on the indicator layer has degraded.

BRIEF DESCRIPTION OF THE DRAWING

Figure 1 shows the indicator in rain in two states: on the left the finish is intact and the indicator is blank; on the right the finish has degraded, the indicator layer is wet and the marking is visible.

DETAILED DESCRIPTION

In one embodiment the indicator layer is a thin, pale woven fabric carrying the same finish as the article, overlying a backing printed with a dark marking such as the word REPROOF. Dry, the pale

layer is opaque and hides the marking. Wet, the pale layer becomes translucent and the marking is visible through it.

While the finish is intact, water beads on the indicator layer and does not wet it, so the marking stays hidden in rain. When the finish has worn away, water wets the layer and the marking appears. On drying the marking is hidden again. After the article and indicator are re-treated with a water-repellent product, the marking again stays hidden in rain.

In another embodiment the marking is concealed by a hydrochromic ink that is opaque when dry and transparent when wet, the ink lying beneath the finished surface.

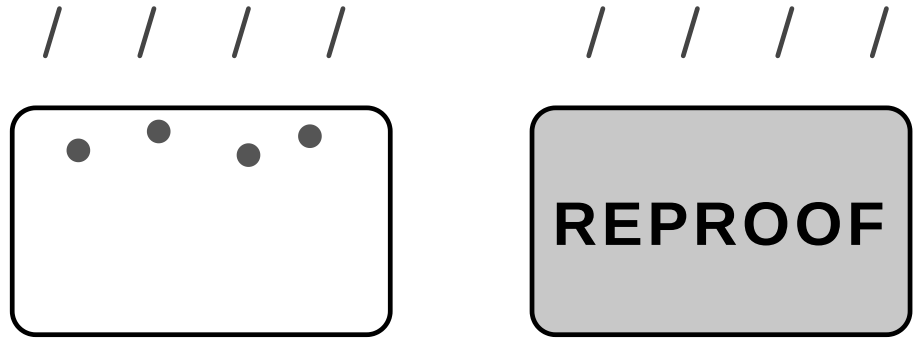
The indicator may be sewn to the article at manufacture and finished in the same bath, or supplied as a separate adhesive or heat-applied patch, for example with a re-treatment product.

VARIATIONS

The following variations are within the scope of the invention, alone or in combination:

1. The indicator layer is the face fabric of the article itself, with the marking on its inner side.
2. The finish on the indicator is selected to fail earlier than the finish on the article, to give advance warning.
3. Two or more markings that are revealed at different stages of wear.
4. Location on a shoulder, cuff, hip belt area, tent fly, awning edge or cover seam.
5. Use on garments, footwear, gloves, tents, swags, awnings, boat covers, outdoor cushions and bags.

The embodiments described are examples only. Features of one embodiment may be combined with features of another.



Coating working
Water beads. Tab stays blank.

Coating worn off
Fabric soaks. The word shows.

Figure 1